Status	Finished
Started	Tuesday, 29 September 2026, 3:50 PM
Completed	Tuesday, 29 September 2026, 3:55 PM
Duration	4 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O👎
- c. O(n log n)
- d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Hashing
- c. Binary search
- d. Dynamic search

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. n/2
- d. log n

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **5**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 100
c. 500
d. 1000

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Hash
c. Binary ✓
d. Linear

Question **7**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 89 and 94
c. 94 and 99
d. 90 and 99 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **11**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is large and unsorted

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the left sublist ✓
c. At the beginning of the list
d. In the entire list

Question **13**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It always starts from the beginning of the list
c. It works on unsorted lists
d. It can be applied only if the list is sorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Both (1) & (3)
c. Hash search
d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search continues until the end of the list
c. The search stops immediately ✓
d. The search backtracks to find duplicate elements